

Matrix multiplication as composition

- last time: what linear transformations are & how to represent them using matrices
 - linear transformations are functions w/ vectors as inputs & outputs
 - can be visualized as smushing around space as long as grid lines parallel & evenly spaced & origin is origin
- $L()$
- linear transformation is completely determined by where it takes the basis vectors of the space, bc any other vector can be described as a linear combination of the basis vectors of the space

• $\vec{v} \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \text{transformed} \rightarrow \begin{bmatrix} x & L(\hat{i}) \\ y & L(\hat{j}) \end{bmatrix}$

• matrix = $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix}$
 where $\hat{i}$ lands where $\hat{j}$ lands

• matrix represents a specific linear transformation

• multiple transformations in a row

$L_1(\)$ followed by $L_2(\) = \text{composition of } L_1 \text{ \& } L_2$

• composition of n lin transforms can also be described by a matrix

• long way of applying both: $\left(\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
 rotation shear

• but same
as just

$$\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

composition
of both

• multiplying two matrices has geometric
meaning of applying one transformation,
then another

← read right to left: $A(g(x))$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

Shear rotation composition

$$\begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix} = ?$$

M_2 M_1

1) where does $\hat{t}$ go?

after 1st transformation,
 $\hat{t}$ goes to $(1, 1)$

after 2nd transformation:

$$\begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

2×2 2×1

$\hat{i}$ after 2nd transform = 1st col of comp. matrix

$$\hat{j}''? \quad \hat{j}' = \begin{bmatrix} -2 \\ 0 \end{bmatrix}, \quad \hat{j}'' = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \end{bmatrix} = -2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

$$\therefore \text{comp matrix} = \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} i_x'' & j_x'' \\ i_y'' & j_y'' \end{bmatrix}$$

generally:

$$\begin{matrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} & \begin{bmatrix} e & f \\ g & h \end{bmatrix} & = & \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix} \\ M_2 & M_1 & & \text{Result} \end{matrix}$$

① follow where $\hat{i}'$ goes. see 1st col of M_1 .

• multiply that column by matrix on the left

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e \\ g \end{bmatrix} = e \begin{bmatrix} a \\ c \end{bmatrix} + g \begin{bmatrix} b \\ d \end{bmatrix} = \text{result 1st col}$$

② follow where $\hat{j}'$ goes - apply 2nd transform for $\hat{j}''$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} f \\ h \end{bmatrix} = f \begin{bmatrix} a \\ c \end{bmatrix} + h \begin{bmatrix} b \\ d \end{bmatrix} = \text{result 2nd col}$$

• matrix multiplication = applying one transform after another.

• order matters: $M_1 M_2 \neq M_2 M_1$
of transforms

• Associativity: $(AB)C \stackrel{?}{=} A(BC)$

There is nothing to prove you're still
applying the same three
linear transformations @
the end of the day

3D linear transformations

• $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \xrightarrow{\text{lin tra}} L(\vec{v}) \rightarrow \begin{bmatrix} e \\ f \\ g \end{bmatrix}$ visualized as shifting 3D grid
in space (grid represents tips of all vectors)

• easier to just think of basis vectors: $\vec{i}, \vec{j}, \vec{k}$

• $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ represents the whole transformation, where each
column = basis vector
after the
transformation

to see
where $\vec{v}$
lands, apply
matrix
as transform: $\begin{bmatrix} r & r & r \\ r & r & r \\ r & r & r \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$